

CO2 ASSESSMENT TOOL 1

SCENARIO-BASED REQUIREMENT ANALYSIS

Project Title

Intelligent E-Governance Citizen Service Request Management System

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Assessment Tool 1 – Scenario-Based Requirement

Analysis

Weightage: 20% | Total Marks: 25

Code of Conduct:

I, [S.Suvikumar / 192424262], certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

1. Problem Statement & Scenario Description

Government departments receive large numbers of citizen service requests related to certificates, public utilities, roads, sanitation, welfare services, grievances, and other civic issues. In conventional processes, requests may be submitted through separate offices, phone calls, paper forms, or disconnected online channels. This can result in delayed responses, duplicate requests, poor visibility of request status, manual workload, and difficulty in measuring service performance.

The Intelligent E-Governance Citizen Service Request Management System is a web and mobile-enabled platform designed to centralize citizen requests, automatically classify and prioritize them, route them to the appropriate department, track service-level deadlines, and provide transparent status updates. The system can use AI-assisted classification and workflow automation to improve response efficiency while maintaining role-based access and auditability.

The system supports three major perspectives: citizens who submit and track requests, government officials who process and resolve requests, and administrators who manage departments, users, workflows, service categories, and reports.

2. Stakeholder Identification & User Classes

User Class / Stakeholder	Role & Responsibilities	Key System Needs
Citizen	Submit service requests, upload supporting documents, track status, receive notifications, and provide feedback.	Simple request form, request ID, status tracking, notifications, document upload, feedback.
Government Service Officer	Review assigned requests, verify information, communicate with citizens, update status, and close resolved requests.	Work queue, request details, SLA alerts, assignment tools, communication and resolution controls.
Department Supervisor	Monitor departmental workload, reassign requests, approve escalations, and review performance.	Department dashboard, workload analytics, escalation controls, SLA monitoring.
System Administrator	Manage users, roles, departments, categories, workflows, configurations, and system health.	Role management, access control, configuration, audit logs, diagnostics.

Government Auditor	Review service records, compliance, response times, and audit trails.	Read-only audit access, historical records, reports, export capability.
Notification Service	Deliver automated SMS, email, and push notifications for important request events.	Reliable APIs, templates, delivery status and retry handling.

3. Requirements Analysis

A. Functional Requirements (FR)

FR-1 (Citizen Registration & Authentication): The system shall allow citizens to register and securely authenticate using approved identity credentials and shall support password recovery and optional multi-factor authentication.

FR-2 (Service Request Submission): The system shall allow citizens to select a service category, enter request details, specify location where relevant, attach supporting documents, and submit a request.

FR-3 (Unique Request Tracking): The system shall generate a unique request ID and allow citizens to view the current status, assigned department, timestamps, and major processing events.

FR-4 (Intelligent Classification & Routing):The system shall classify requests using category, keywords, location, and configurable rules or AI-assisted classification, then route them to the appropriate department or officer.

FR-5 (Priority & SLA Management):The system shall assign configurable priorities and service-level deadlines and shall alert responsible officers when deadlines approach or are breached.

FR-6 (Officer Processing):The system shall allow authorized officers to accept, review, request additional information, update status, add remarks, attach resolution evidence, and close requests.

FR-7 (Notifications):The system shall send notifications through configured channels such as SMS, email, or push notifications when requests are submitted, assigned, updated, escalated, or resolved.

FR-8 (Escalation Management):The system shall automatically or manually escalate overdue or critical requests to supervisors according to configured escalation rules.

FR-9 (Feedback & Rating):The system shall allow citizens to provide feedback and service ratings after a request is resolved.

FR-10 (Analytics & Reporting):The system shall generate dashboards and downloadable reports showing request volume, categories, resolution time, SLA compliance, departmental workload, and citizen feedback.

FR-11 (Role-Based Access Control):The system shall authenticate users and restrict system privileges according to roles such as Citizen, Officer, Supervisor, Administrator, and Auditor.

FR-12 (Audit Trail):The system shall record significant actions, status changes, assignments, approvals, escalations, and administrative changes for accountability.

B. Non-Functional Requirements (NFR)

NFR-1 (Performance & Scalability):The system shall support high volumes of concurrent citizen requests and provide normal dashboard and request operations with a target response time of less than 3 seconds under expected load.

NFR-2 (Availability & Reliability):The production platform shall target 99.9% availability and shall use backup and recovery mechanisms to reduce service disruption.

NFR-3 (Security & Data Protection):Sensitive data shall be protected using encryption in transit and at rest, secure authentication, role-based authorization, session controls, and audit logging.

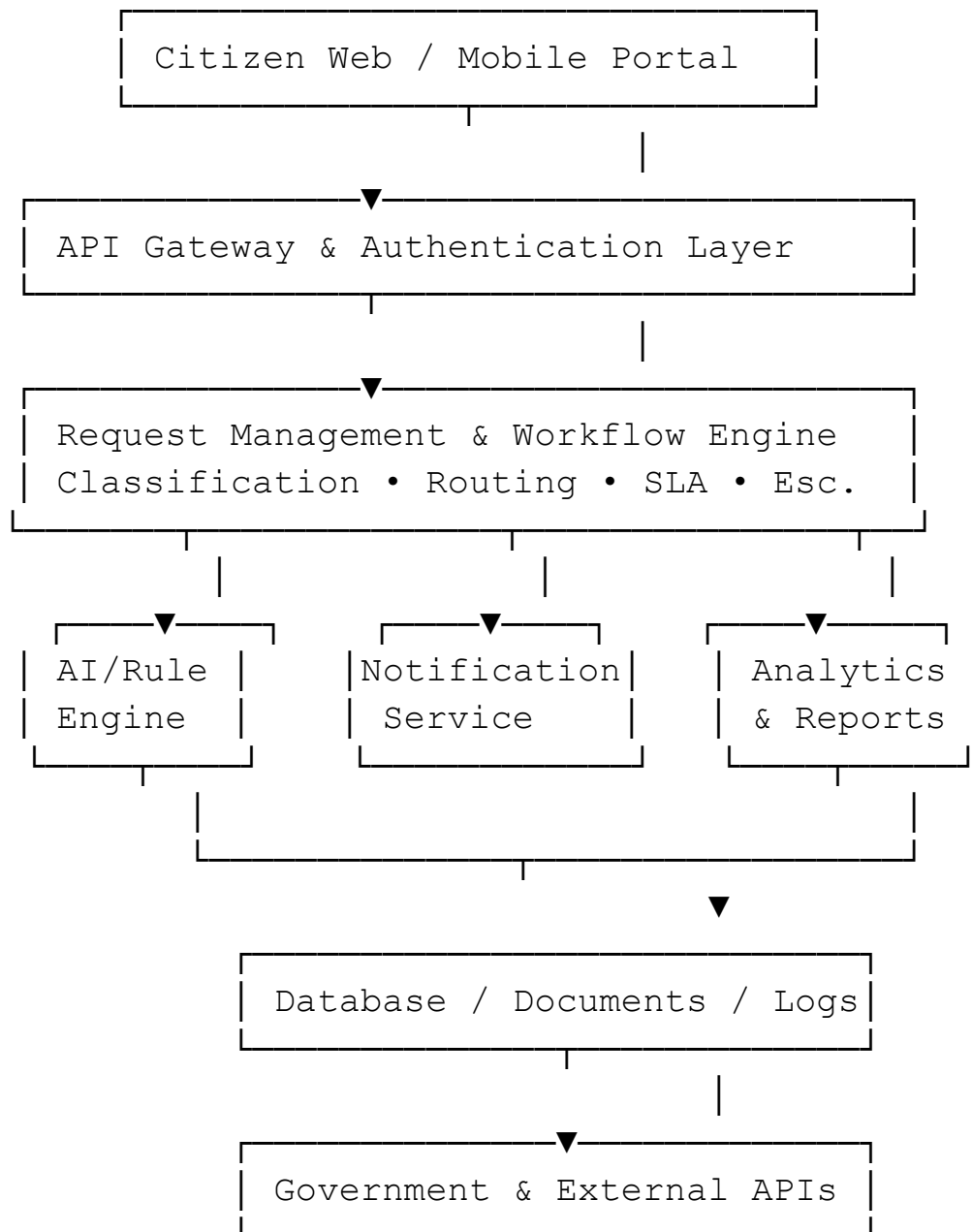
NFR-4 (Usability & Accessibility):Citizen-facing web and mobile interfaces shall be intuitive, responsive, and designed with accessibility principles consistent with WCAG 2.1 Level AA.

NFR-5 (Maintainability):The system shall use modular components, documented APIs, logging, testing, and configuration-driven workflows to support maintenance and future enhancements.

NFR-6 (Interoperability):The platform shall provide secure APIs for integration with identity services, departmental systems, notification gateways, mapping services, and other approved government platforms.

4. System Architecture & High-Level Block Diagram

The proposed architecture follows a layered, service-oriented approach. Citizens and government personnel access the system through web or mobile interfaces. An application/API layer manages authentication, request processing, workflow, notification, analytics, and AI-assisted classification. A data layer stores request records, user information, documents, audit logs, and reporting data. External services provide identity verification, SMS/email/push notifications, maps, and departmental integrations.



5. Prioritisation Matrix (MoSCoW Method)

Requirement	Description	Category	Justification
FR-1	Citizen authentication and registration	Must Have	Required for secure access and personalized service.
FR-2	Citizen service request submission	Must Have	Core function of the platform.
FR-3	Request ID and status tracking	Must Have	Provides transparency and reduces repeated enquiries.
FR-4	Intelligent classification and department routing	Must Have	Reduces manual routing and improves processing speed.
FR-5	Priority and SLA management	Must Have	Supports timely government service delivery.
FR-6	Officer request processing and resolution	Must Have	Essential for completing service requests.
FR-7	Automated notifications	Should Have	Keeps citizens and officials informed.
FR-8	Escalation management	Should Have	Helps prevent unresolved and overdue requests.
FR-9	Citizen feedback and rating	Could Have	Useful for service quality improvement.
FR-10	Analytics and management reports	Could Have	Supports performance monitoring and planning.
FR-11	Role-based access control	Must Have	Protects functions and data from unauthorized access.
FR-12	Audit trail	Should Have	Improves accountability and traceability.

Future	Predictive demand forecasting and advanced generative-AI assistance	Won't Have (Now)	Can be introduced after the core workflow is stable and sufficient data is available.
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6. Use Case Summary

Actor	Main Use Cases
Citizen	Register/Login; Browse Services; Submit Request; Upload Documents; Track Status; Receive Notifications; Provide Feedback.
Service Officer	Login; View Assigned Requests; Verify Details; Update Status; Request Information; Resolve Request; Add Remarks.
Supervisor	View Department Queue; Reassign Requests; Monitor SLA; Approve Escalations; Review Performance.
Administrator	Manage Users/Roles; Manage Departments; Configure Services; Configure Workflow/SLA; View Audit Logs.
Auditor	Search Records; Review Audit Trail; Generate Compliance/Performance Reports; Export Data.

7. Key Data Entities

Entity	Important Attributes
Citizen	Citizen ID, name, contact details, authentication information, preferred notification channel.
Service Request	Request ID, category, description, location, priority, status, created date, SLA deadline, resolution date.
Department	Department ID, name, service categories, contact information.
Officer	Officer ID, name, department, role, status.

Attachment	Attachment ID, request ID, filename, type, storage reference, upload timestamp.
Notification	Notification ID, request ID, channel, message template, delivery status, timestamp.
Feedback	Feedback ID, request ID, rating, comments, submitted date.
Audit Log	Log ID, actor, action, affected record, timestamp, result.

8. Main Workflow

1. Citizen logs into the portal and selects the required government service.
2. Citizen enters the request information and uploads supporting documents if required.
3. The system validates mandatory fields and generates a unique request ID.
4. The classification and routing component determines the appropriate category and department.
5. The request is placed in the responsible officer's queue with priority and SLA deadline.
6. The officer reviews the request, communicates with the citizen if clarification is required, and updates the status.
7. The system sends status notifications and monitors SLA deadlines.
8. Overdue or critical requests are escalated according to configured rules.
9. After resolution, the citizen receives a completion notification and may provide feedback.
10. Management dashboards and audit records are updated for monitoring and reporting.

AI-Powered Smart Aquaculture Platform



9. Assumptions, Constraints & Risks

Assumptions

- Citizens and authorized government staff have access to supported web or mobile devices.
- Government departments provide accurate service categories, routing rules, and SLA definitions.
- Notification and identity services are available through approved secure interfaces.
- Departmental officers are trained to use the workflow and update request status accurately.

Constraints

- The platform must comply with applicable government security, privacy, accessibility, and records-management requirements.
- Integration quality may depend on the availability and standards of existing departmental systems.
- AI-assisted classification should not replace authorized human decisions for sensitive or exceptional cases.

Major Risks & Mitigation

Risk	Impact	Mitigation
Incorrect request classification	Request routed to wrong department	Confidence thresholds, rules, human correction and retraining.
Unauthorized access	Data privacy and security breach	MFA, RBAC, encryption, logging and periodic access review.
Notification failure	Citizen misses important update	Retry queues, delivery tracking and alternate channels.
SLA data misconfiguration	Incorrect escalation or reporting	Central configuration, validation and supervisor review.
System outage	Service disruption	Monitoring, backups, disaster recovery and graceful error handling.

10. Summary & Cross-Disciplinary Integration

The Intelligent E-Governance Citizen Service Request Management System integrates software engineering, database management, web and mobile application development, cloud computing, secure API integration, workflow automation, data analytics, and responsible AI techniques. The system addresses a practical governance problem by creating a centralized lifecycle for citizen service requests—from submission and classification to assignment, monitoring, resolution, feedback, and audit.

The requirements analysis identifies the stakeholders, functional and non-functional requirements, architecture, workflow, and MoSCoW priorities needed for a software engineering project. The proposed solution emphasizes transparency, accountability, timely service delivery, secure access, and measurable departmental performance. Advanced predictive analytics and more sophisticated AI features can be introduced in future iterations after the core request-management workflow is stable.

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